Multi-digit addition: one model from many

Recognition, associations, problem solving, and memory, composed and proved

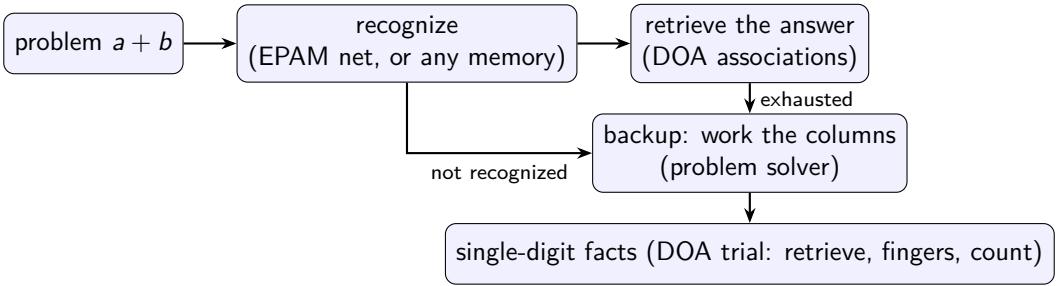
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The task, and what a model must explain

- Children and adults solve problems like $150 + 100$ and $478 + 256$.
- Some problems are *familiar*: the answer is retrieved. Others are worked out, column by column.
- Errors come in kinds:
  - **confident errors**: a familiar-looking problem answered with a neighbour's sum, with no hesitation;
  - **column errors**: a single-digit fact wrong in one column;
  - **procedural bugs**: systematic errors from a procedure with a missing step (no carry).

The architecture



Every arrow is a composition of probability distributions; every box is a theory formalized on its own, with its own theorems.

Component 1: recognition (EPAM)

- A **discrimination net** tests features of the problem's code, and sorts it to a stored image.
- Learning adds a test only where two codes are confused.
- A net that has never needed to tell two problems apart can't: $150 + 100$ is sorted to $100 + 100$ when the net tests only the digit where the familiar problems differ.

Proved for every net: recognition depends only on the features tested; a confusion agrees on every tested feature; learning a problem never disturbs another.

Component 2: distributions of associations (Siegler & Shrager 1984)

- Each problem has associations of varying strength to possible answers.
- Retrieval samples an answer in proportion to strength; it is stated if above a **confidence criterion**; up to a search length.
- If nothing is stated: backup strategies (fingers, counting fingers).
- Used twice here: for **whole problems** (once recognized) and for **single-digit facts** in each column.

Proved: retrieval is used with probability $1 - (1 - R)^n$; the error share on retrieval trials is the wrong answers' share above the criterion; strengthening the correct answer never raises errors.

Component 3: the column backup as problem solving

- A problem space: add a column (its fact from a DOA trial), write, write and carry, finish.
- **The bridge**: the solver answers exactly as the old fixed column procedure did, for every problem and every fact process.
- **Means-ends analysis** (GPS) works units first, though its goal names no order.
- **Repair theory** (Brown & VanLehn 1980): delete the carry rule, and a column sum of 10 or more is an impasse; its repairs are bugs (no carry: $15 + 17 = 22$).

Component 4: memory as a parameter

- Recognition is any **retriever**: study, then retrieve a distribution over stored items.
- EPAM's net is one; ACT-R's declarative memory (activation, partial matching) is another.
- Three **laws** of memory, determinacy, certainty and stability, are enough for confident errors; EPAM's net has all three, ACT-R has none.

So which errors a learner makes depends on which memory they have, and the laws say why.

How they compose

A trial on problem x:

$$\text{trial}(x) = \text{recognize}(x) \gg\!= (\text{retrieve the recognized problem's answer else backup}(x))$$

$$\text{backup}(x) = \text{add units} \gg\!= \text{write and carry} \gg\!= \text{add tens} \gg\!= \ldots$$

$$\text{add column} = \text{DOA trial on its digits} \gg\!= \text{record the sum}$$

- Each $\gg\!=$ is a two-stage draw: probabilities multiply along a path and add over paths.
- Composition is exact: no simulation anywhere; every number is a theorem.
- Because draws regroup (associativity), a theorem about a component holds inside the whole.

An application built from the library

Layer	What it supplies	Where
Library	discrimination nets and their learning; sampled retrieval with a criterion; recognize, then retrieve, else back up; the decision cycle with impasses and repairs; exact distributions	`CogLib/`
Theory	the single-digit trial: retrieve, fingers, count (Siegler & Shrager)	`StrategyChoice/`
Task knowledge	the problem's code; the column problem space and its procedure	`MultiDigit/`
The application	the wiring: the solver as the backup, a DOA trial as each column's fact	`Integrated.lean`

- No new mechanism, and no answer given: correct knowledge enters only as single-digit associations and the carry rule.
- It inherits every theorem of its parts, and proves its own about the whole.

What rests on what

Kind	Results
Follows from the composition, at every setting	the solver answers as the column backup; whole-problem errors are untouched by the solver, so a better solver cannot repair them; a solver error is a single-digit error moved to its column
Given stated conditions	confident errors, and two sources of error on a confused problem, given: the confusion (derived from the learned net), sharp associations (assumed, not yet explained), positive strengths (well-formedness)
Illustrative numbers	the worked case, next

No phenomenon is a premise of any of these theorems.

What the whole predicts

Prediction	From
Confident errors: every retrieval of a confused problem is wrong, with no extra backup	recognition (EPAM laws) + sharp associations (DOA)
A column error is a single-digit error moved to its place	problem solver + DOA facts
Carrying adds no error of its own	problem solver (bridge)
Procedural bugs only on carry problems	problem solver without the carry rule
Learning removes a confusion without disturbing other problems	EPAM learning

A worked case

	EPAM	ACT-R
$150 + 100$ answered 200 (neighbour's sum) by retrieval	45/46	225/529
$150 + 100$ reaches the solver	1/46	383/1058
$150 + 100$ answered correctly, by the solver	$\approx .014$	$\approx .238$
$15 + 17$, no carry rule, strict critics	22, for every such learner	

- Values are exact, at settings chosen for illustration; the predictions above hold for every setting. Each column's facts follow one chosen rule.
- The solver alone on $150 + 100$: 250 $\approx .66$; 260, 350, 251 $\approx .10$ each, single-digit errors in place (tens, hundreds, units).
- With EPAM, the confusion is certain; with ACT-R, it is graded.

Is the formalization faithful?

- A prose specification, with no Lean in it, and preregistered predictions.
- A fresh AI agent, given only the prose, wrote a Python simulator.
- **No failures**: 243 outcome frequencies on 4 instances matched Lean's exact values; every applicable prediction passed.
- It also found five vacuous predictions and a gap in the formalization, since fixed or recorded.

The blind check predates the problem solver; by the bridge, its results carry over unchanged.

What each component contributes

Component	Source	Role in the whole
Discrimination nets	EPAM (Feigenbaum; Simon)	which problems look alike; certain confusions
Distributions of associations	Siegler & Shrager 1984	answers, confidence, backup choice; single-digit facts
Problem solving	GPS; Brown & VanLehn 1980	the column procedure; its order; its bugs
Retrievers and laws	ACT-R (Anderson); laws new here (the theory matrix)	which memory gives which errors, and why

Each was formalized and proved on its own; the composition is proved too.

Limits, and next

- Worked cases use chosen settings; priors, whole-problem and single-digit, should come from experience, as Siegler & Shrager's did from counting (open issue).
- "Multi-digit" is three-digit.
- **Chunking** would close the loop: a fact found by counting at a retrieval impasse becomes an association, DOA's learning rule as problem-solving learning.
- Subtraction, borrowing, and repair theory's own bug data.